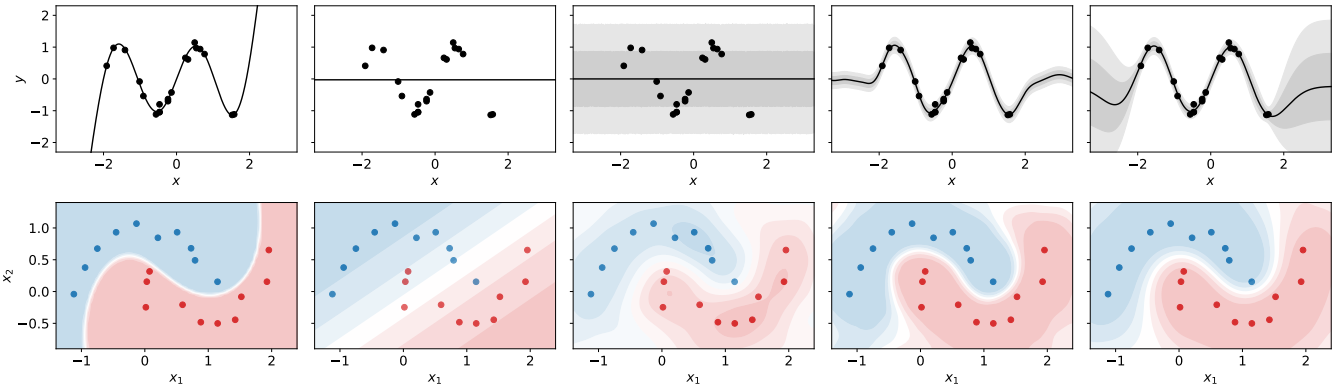

(a) RFF ($\gamma = 1$) MAP + GS (b) RFF ($\gamma = 20$) MAP + GS (c) RFF (learned γ, ν) ELBo (d) RFF (learned γ, ν) DE ELBo (ours) (e) GP (learned γ, ν) Marginal likelihood

Figure 2: **Case Study A: Demo of hyperparameter sensitivity and selection for RFF models.** The first four columns use the RFF regression/classification model in Sec. 3.1, varying estimation and selection techniques. The last column shows the reference fit of a GP, a gold-standard for this toy data but less scalable. Columns a and b show MAP point estimation, with fixed γ and grid search (GS) for ν (separate GD run for each value). Columns c and d use variational methods, *learning* γ, ν in one run of GD. For the regression dataset, we plot the predictive posterior $p(y_*|y_{1:N})$. Our DE ELBo objective enables the best approximation of the GP.